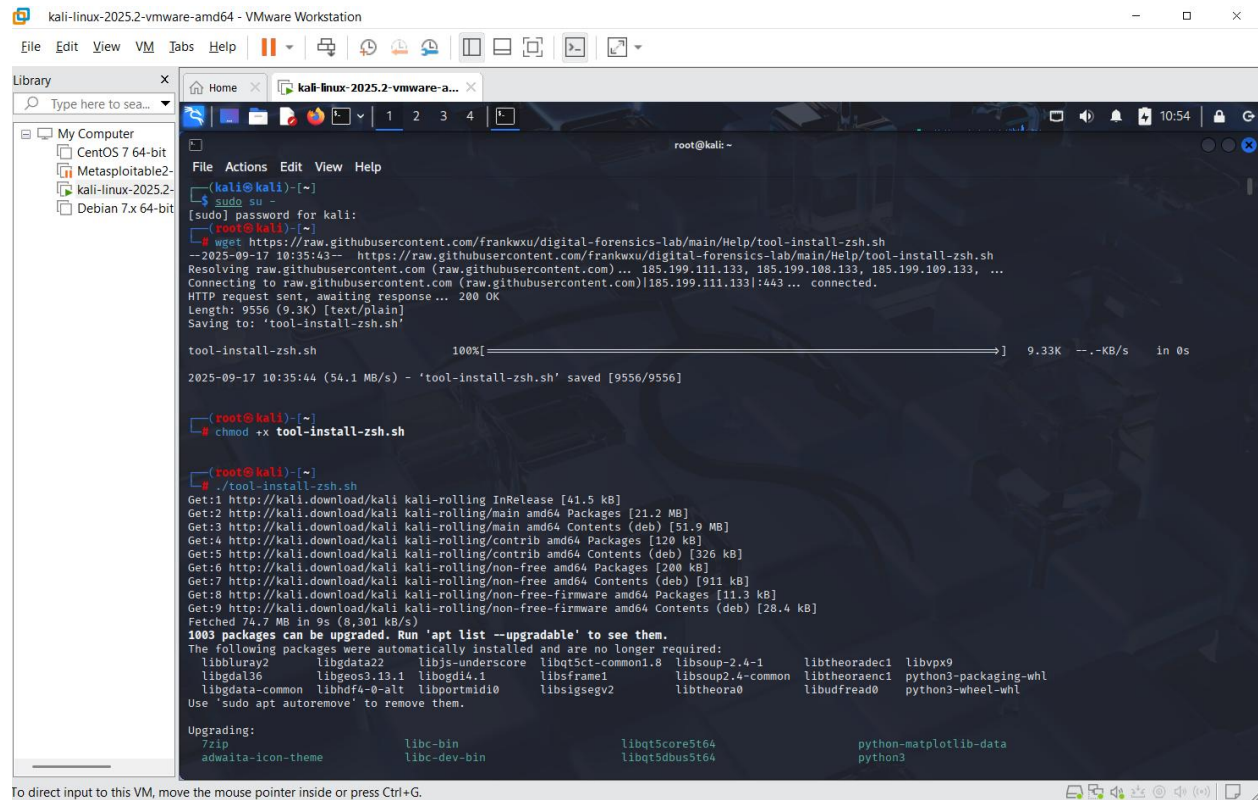


Lab 2: Data Carving

Scenario 1:

Step 1:



```
kali@kali:~$ sudo su -
[sudo] password for kali:
root@kali:~# wget https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Help/tool-install-zsh.sh
--2025-09-17 10:35:43-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Help/tool-install-zsh.sh
Resolving raw.githubusercontent.com (raw.githubusercontent.com) ... 185.199.111.133, 185.199.106.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9556 (9.3K) [text/plain]
Saving to: 'tool-install-zsh.sh'

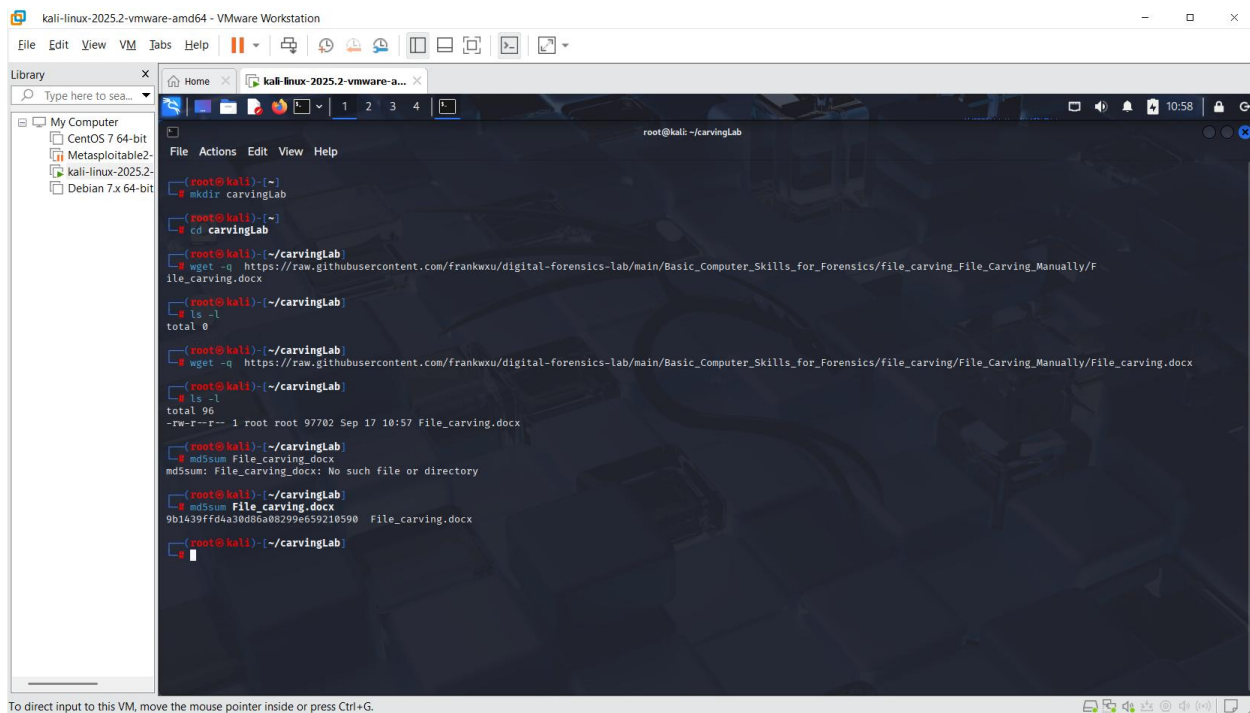
tool-install-zsh.sh      100%[=====] 9.33K  --.-KB/s  in 0s

2025-09-17 10:35:44 (54.1 MB/s) - 'tool-install-zsh.sh' saved [9556/9556]

root@kali:~# chmod +x tool-install-zsh.sh

root@kali:~# ./tool-install-zsh.sh
Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.2 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [51.9 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [120 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [326 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [200 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [911 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [11.3 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [28.4 kB]
Fetched 74.7 MB in 9s (8,301 KB/s)
1003 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
libbluray2 libdata22 libjs-underscore libqt5ct-common1.8 libsoup-2.4-1 libtheoradec1 libvpx9
libgdal36 libgeos3.13.1 libgdi4.1 libframe1 libsoup2.4-common libtheoraenc1 python3-packaging-whl
libgdata-common libhdf4-0-alt libportmidi0 libsigsegv2 libtheora0 libudfread0 python3-wheel-whl
Use 'sudo apt autoremove' to remove them.

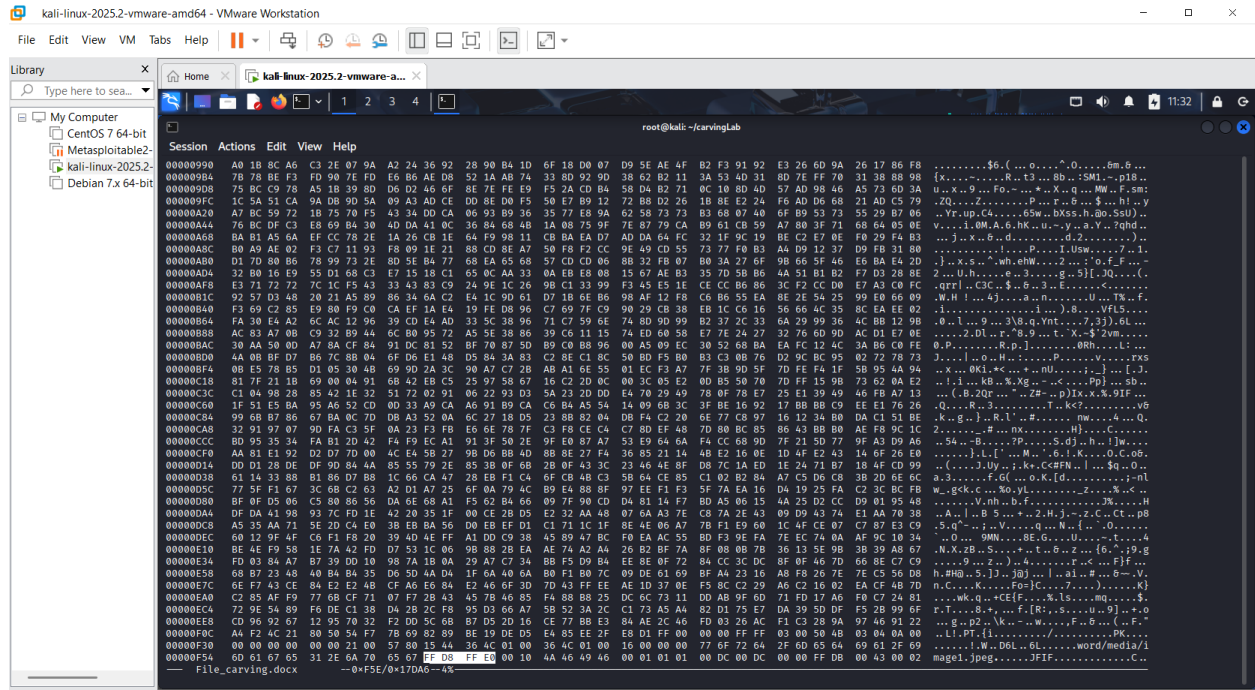
Upgrading:
7zip libc-bin libqt5core5t64 python-matplotlib-data
adwaita-icon-theme libc-dev-bin libqt5dbus5t64 python3
```



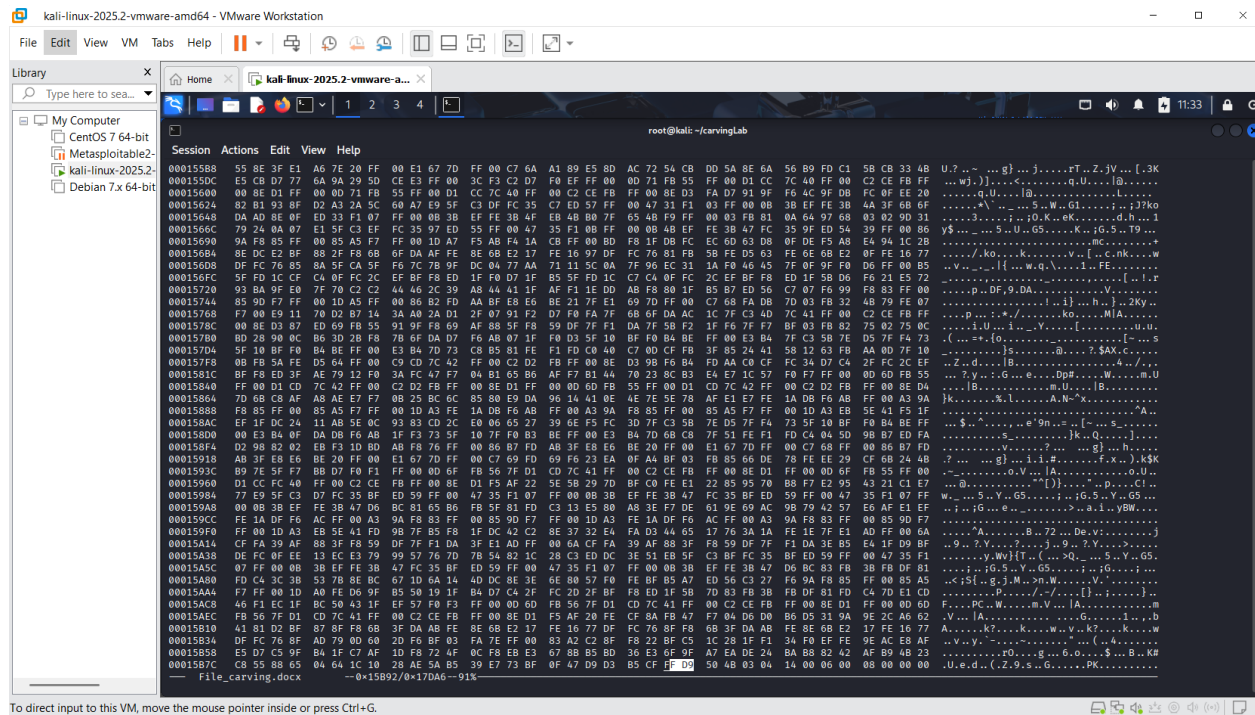
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2 ảnh trên: chuẩn bị file và công cụ yêu cầu

Step 2.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2 ảnh trên: mở file và xem mã hex bằng *hexedit File_carving.docx* sau đó tìm và xác định header / trailer

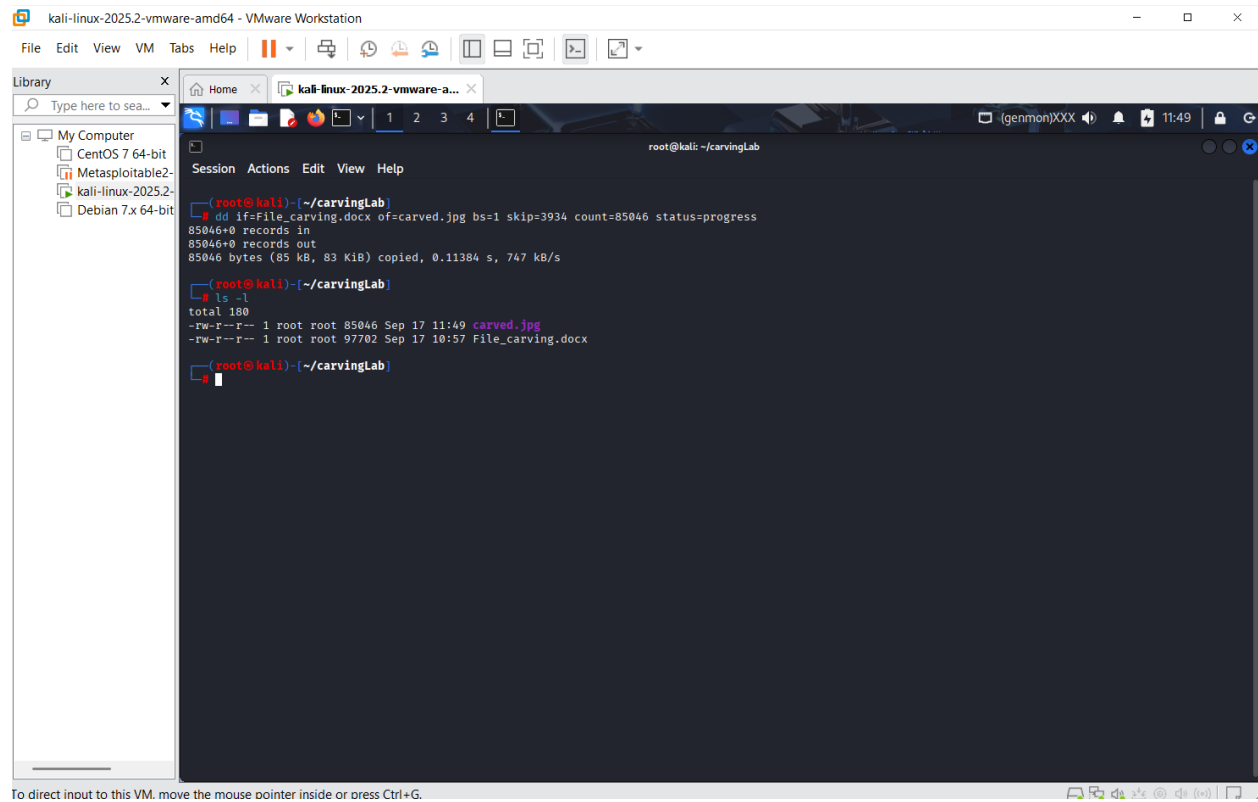
Ta có:

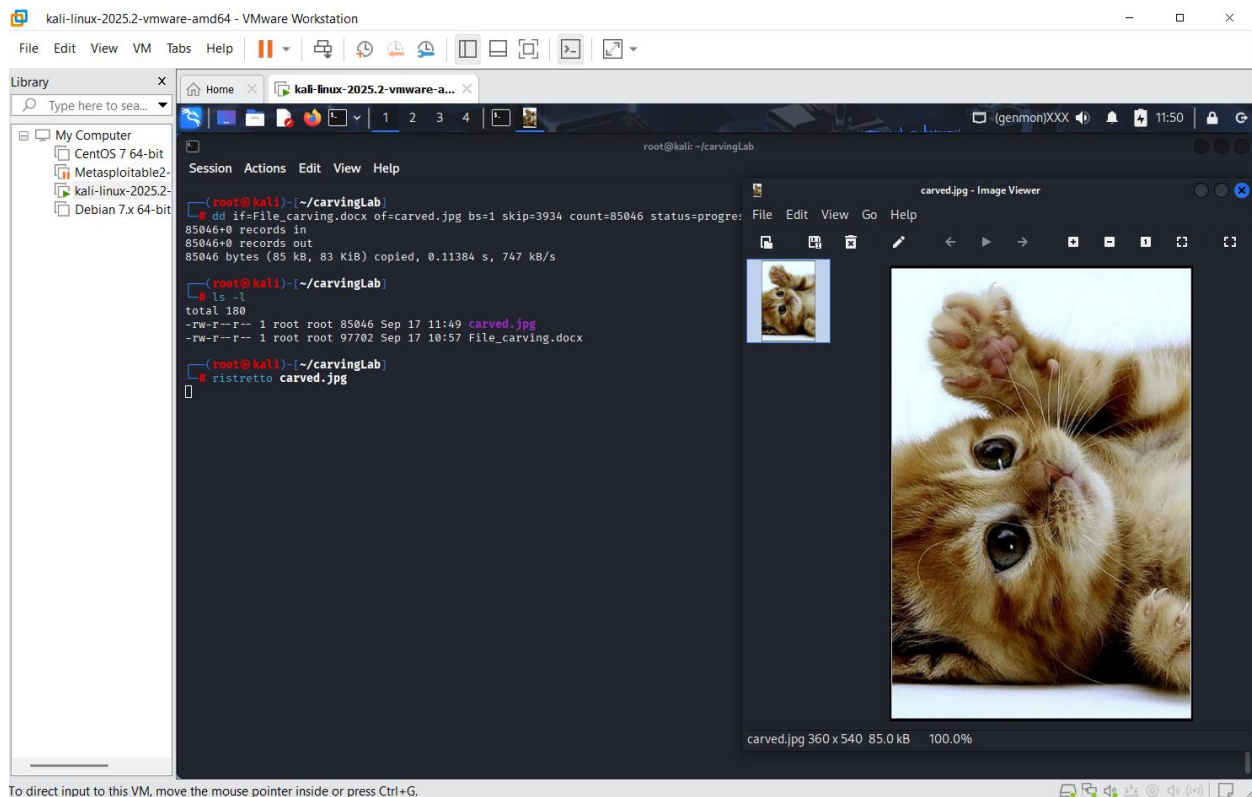
• (0F5E)16=(3934)10

• (15B93)16=(88979)10

count=<end_offset - start_offset + 1>

start = 3934 end = 88979 count = 85046





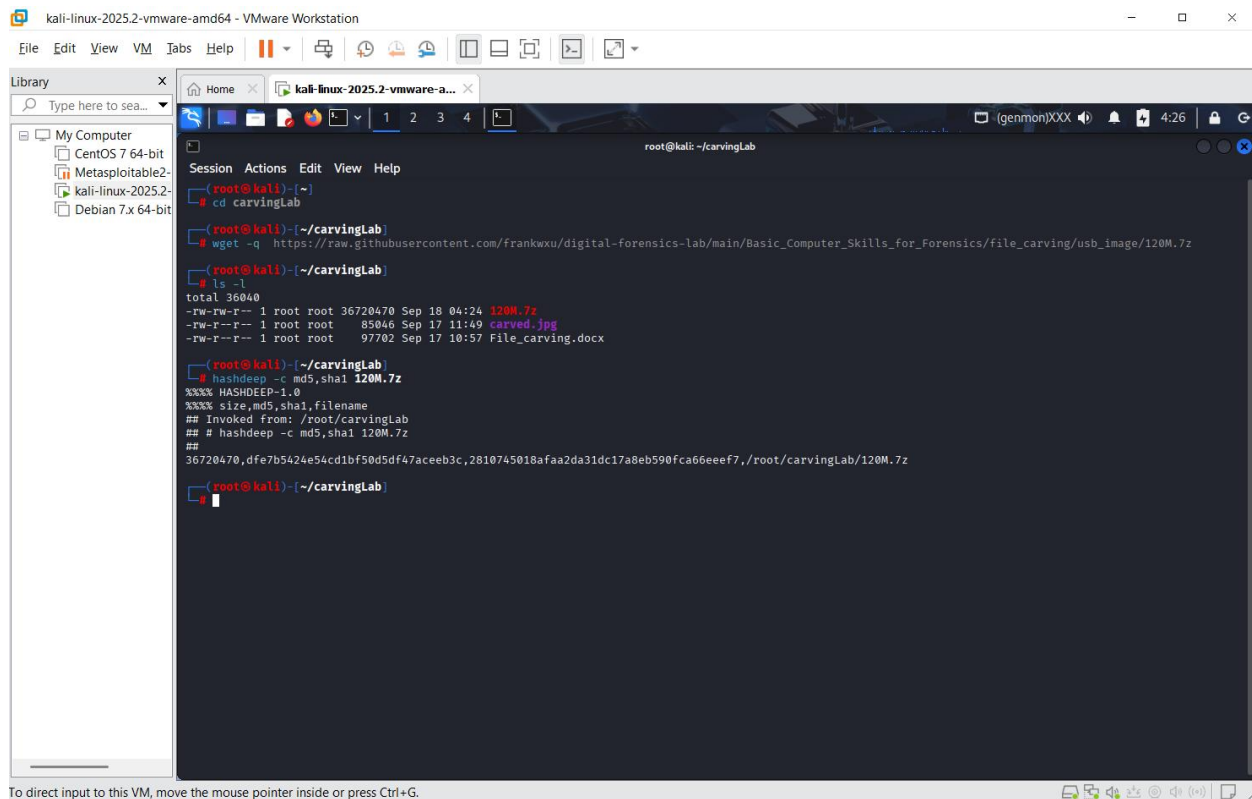
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

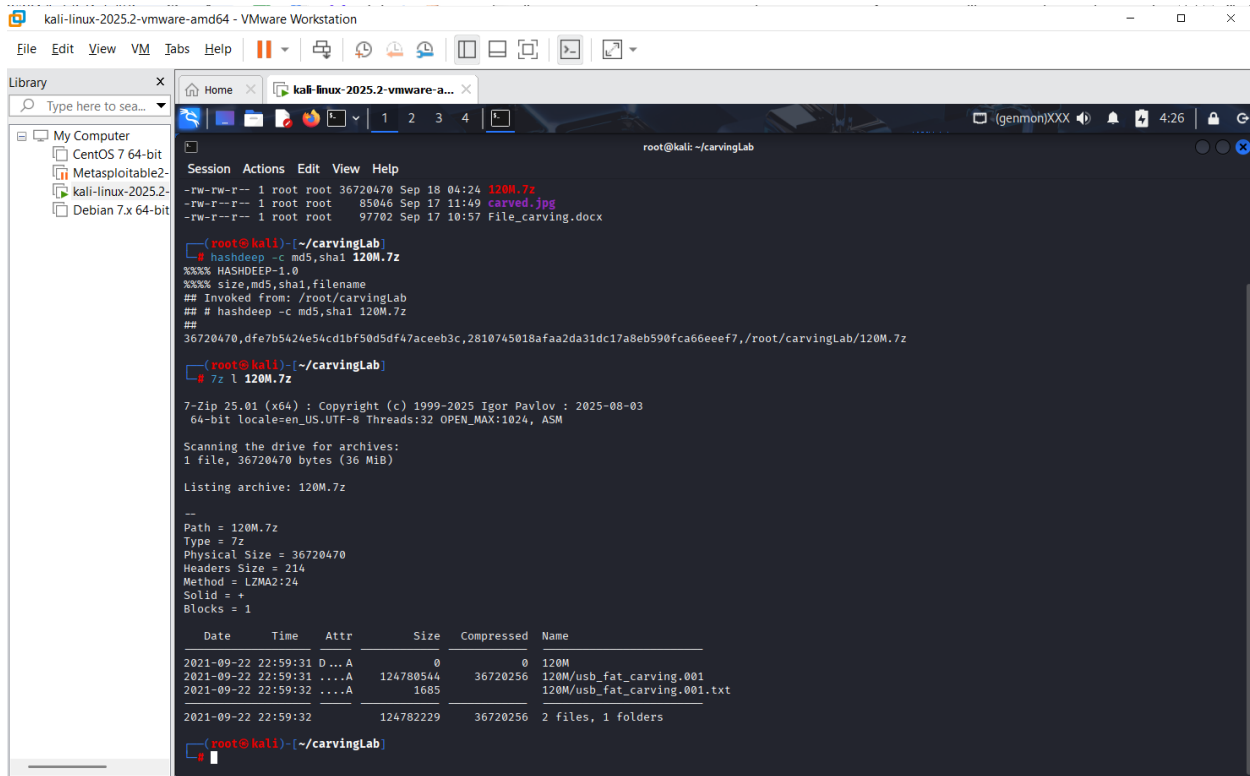
2 ảnh trên: carve file dùng lệnh **dd if=File_carving.docx of=carved.jpg bs=1 skip=3934 count=85046 status=progress** để copy đoạn mã hex ra file carved.jpg. Sau đó dùng lệnh **ristretto carved.jpg #XFCE** viewer để xem ảnh

Scenario 2:

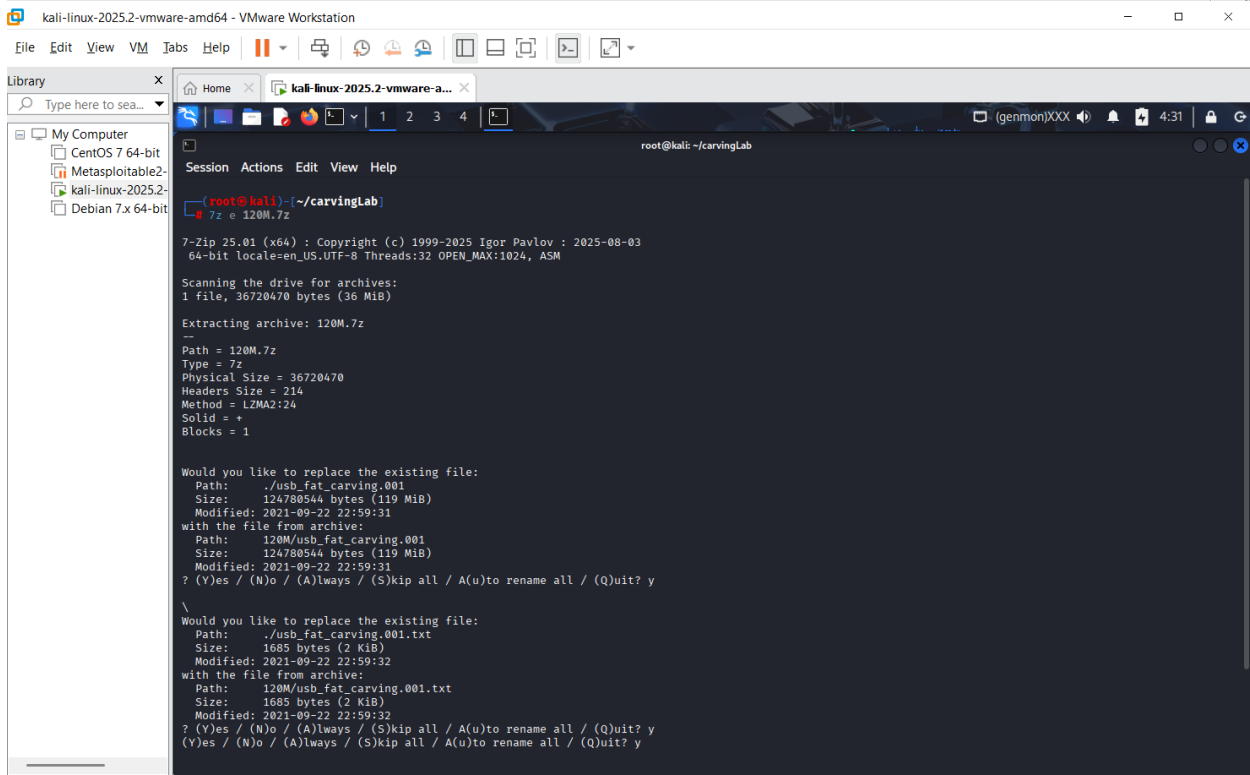
2. Carving/Recovering a USB image

Step 1

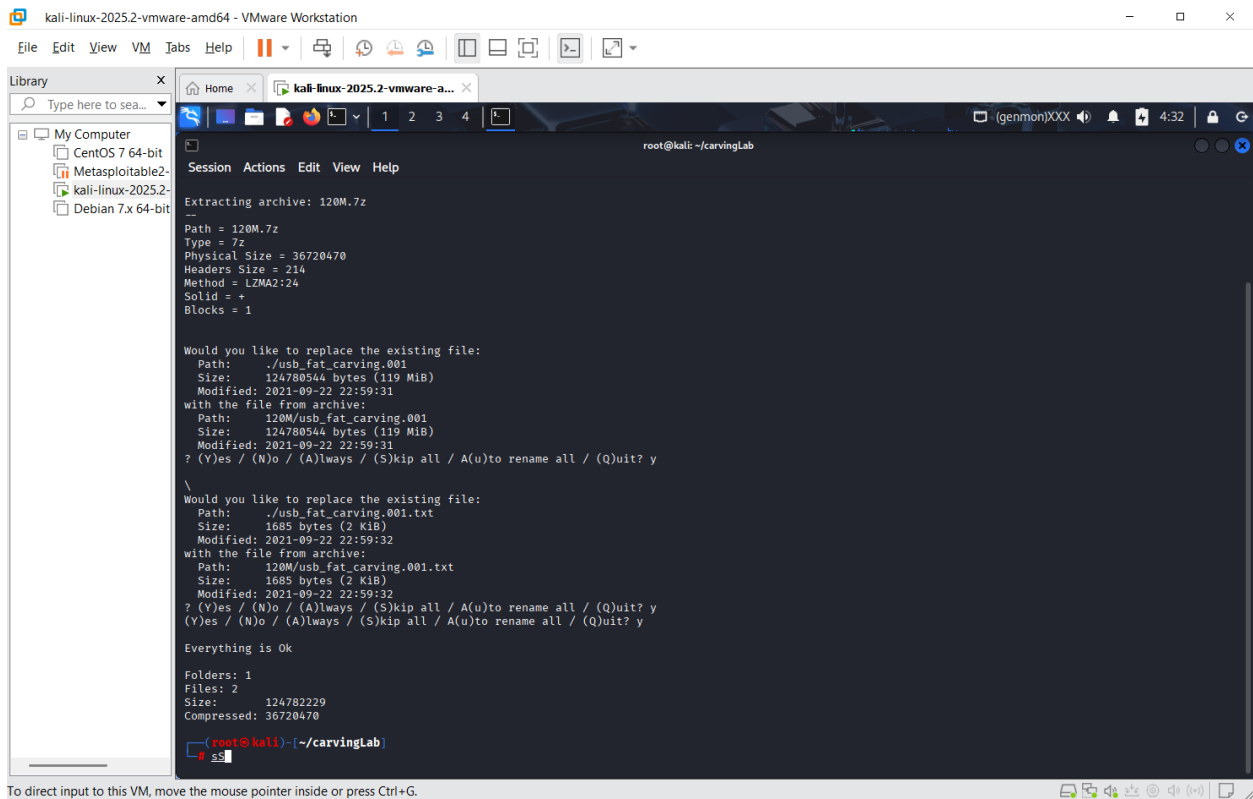




To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

4 ảnh trên: Download the zipped USB image và List the content of the zipped file


```
root@kali: ~/carvingLab
root@kali:~# cat usb_fat_carving.001
#####
##### size,md5,sha256,filename
#####
## Invoked from: /root/carvingLab
## # hashdeep c md5,sha1 usb_fat_carving.001
##
12a780544,ba4a1d0ba9f4a6667b0ba3b3e85e604,9bfe4b563ade38764f0e581a4686938f7fab0472378595b789bcb0248c91d9,./root/carvingLab/usb_fat_carving.001

root@kali:~# cat usb_fat_carving.001.txt grep 'checksum'
Created By AccessData® FTK® Imager 4.5.0.3

Case Information:
Acquired using: ADT-5.0.3
Case Number: USB File Carving
Evidence Number: UB-002
Unique description: USB files used for file carving
Examiner: Frank Xu
Notes: University of Baltimore

Information for D:\cloud_storage\OneDrive - University of Baltimore\UBal\research\DOJ\DOJ_Team_Repository\DOJ_Case_Study\0_Basic_Computer_Skills_for_Forensi
cs\File_carving\usb_image\120M\usb_fat_carving:

Physical Evidentiary Item (Source) Information:
[Device Info]
Source Type: Physical
[Drive Geometry]
Cylinders: 15
Tracks per Cylinder: 255
Sectors per Track: 63
Bytes per Sector: 512
Sector Count: 243,712
[Physical Drive Information]
Drive Model: General UDisk USB Device
Drive Serial Number:
Drive Interface Type: USB
Removable drive: True
Source data size: 119 MB
Sector count: 243712
[Computed Hashes]
MD5 checksum: ba4a1d0ba9f4a6667b0ba3b3e85e604
SHA1 checksum: bcc2d49fd49c9521ecb1739f6542cbbf327375ef

Image Information:
Acquisition started: Wed Sep 22 22:59:16 2021
Acquisition finished: Wed Sep 22 22:59:31 2021
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

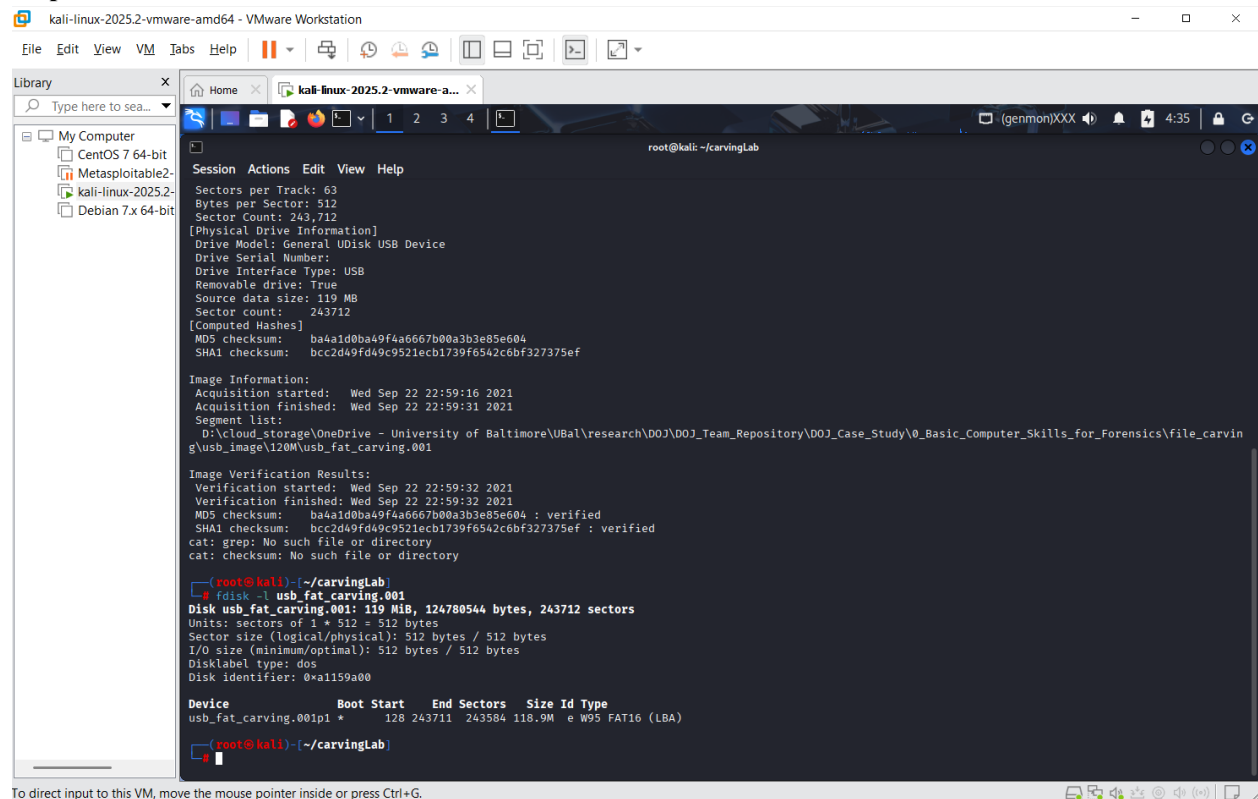
```
root@kali: ~/carvingLab
root@kali:~# cat: grep: No such file or directory
cat: checksum: No such file or directory

root@kali:~#
```

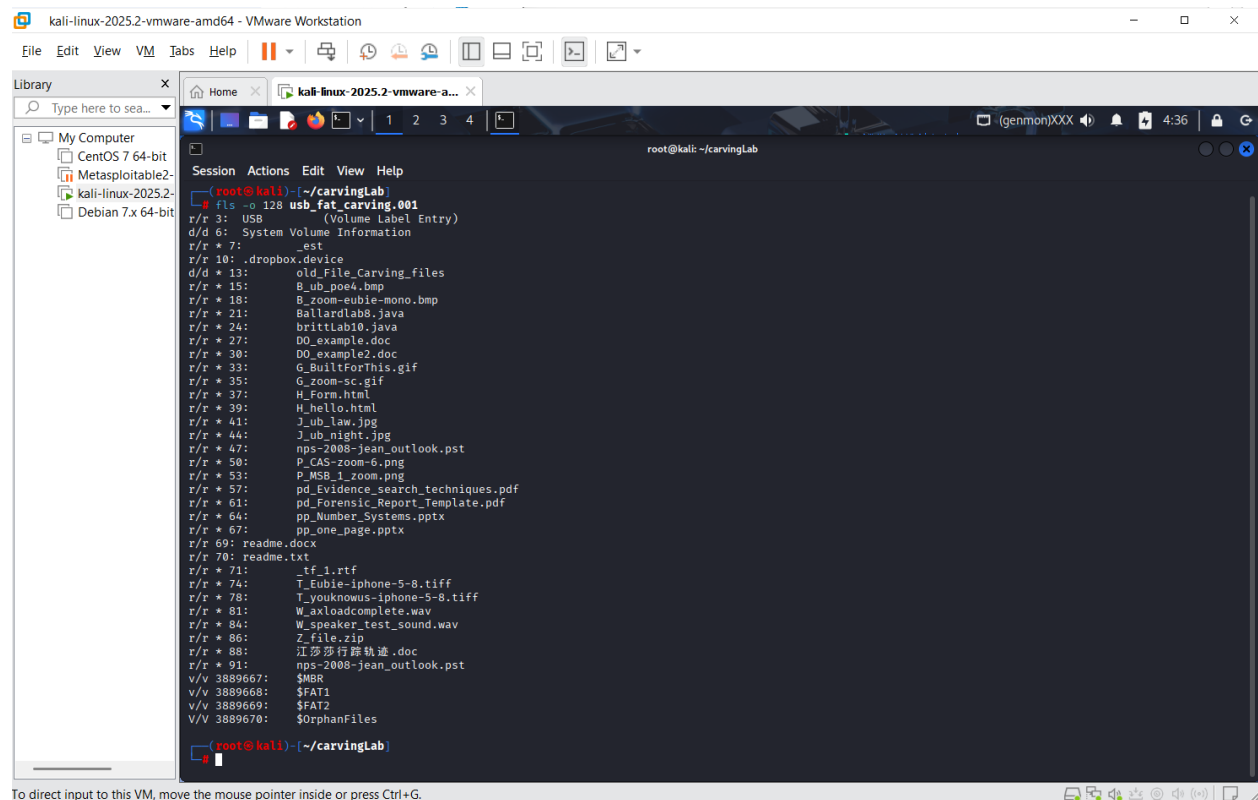
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2 ảnh trên: Xác minh các giá trị băm

Step 2:

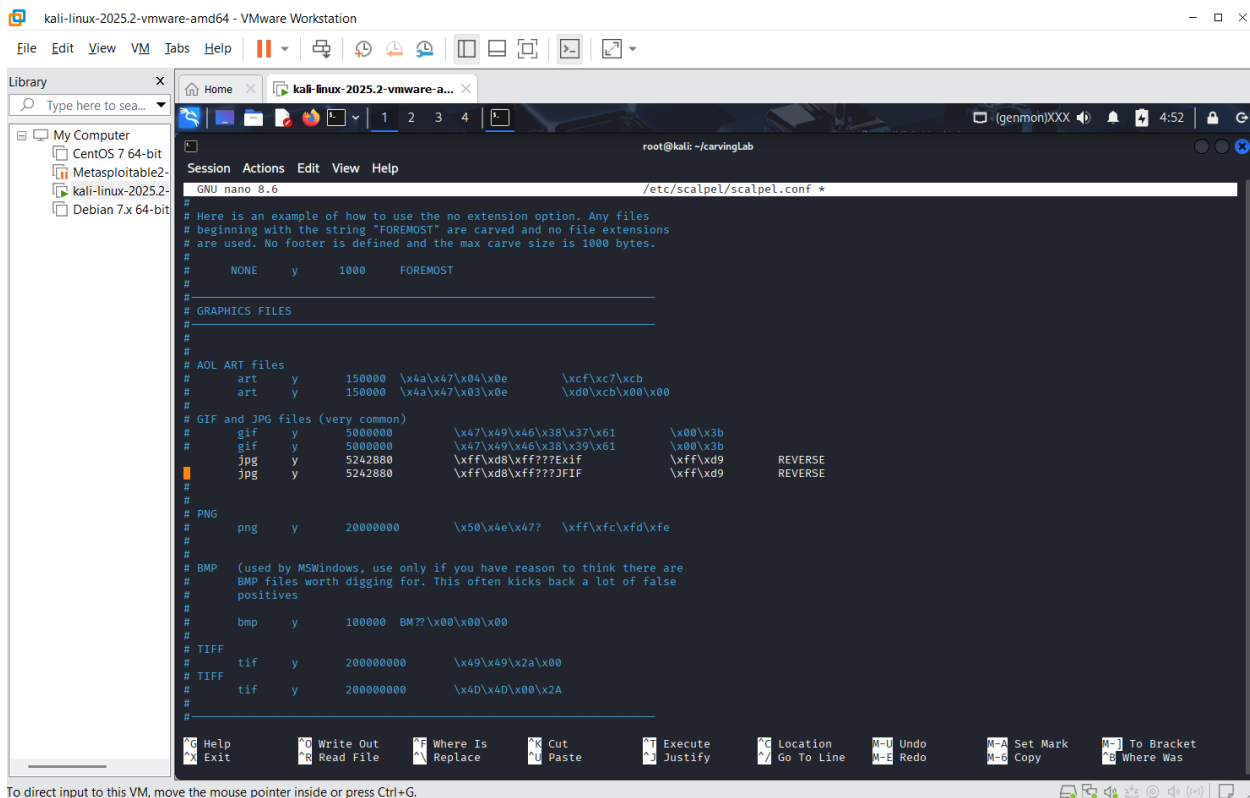


To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

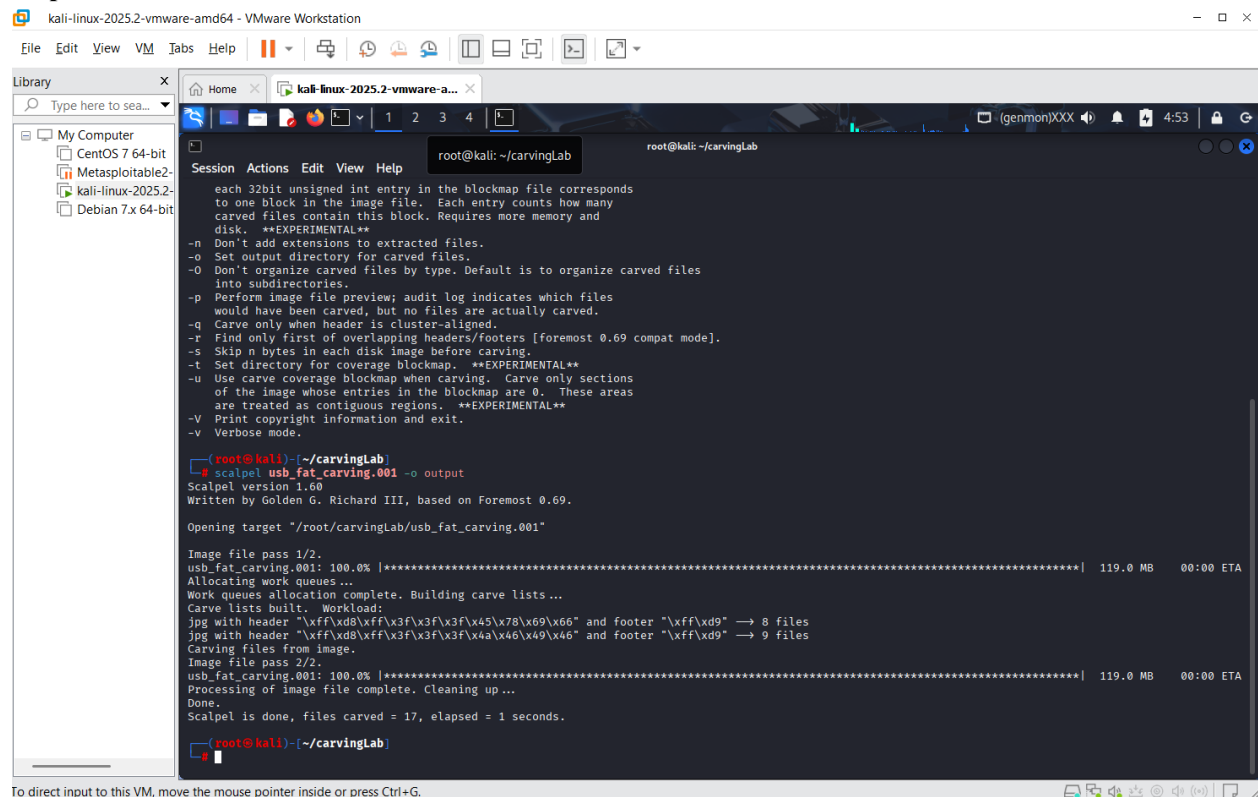
2 ảnh trên: kiểm tra nội dung của USB và hiển thị các phân vùng



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

ảnh trên: Quyết định loại tệp nào cần carve bỏ dấu # ở đầu dòng

Step 3.



```
kali-linux-2025.2-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

Library
Type here to search...
My Computer
CentOS 7 64-bit
Metasploitable2
kali-linux-2025.2-
Debian 7.x 64-bit

kali-linux-2025.2-vmware-a...
1 2 3 4

root@kali: ~/carvingLab
Session Actions Edit View Help

each 32bit unsigned int entry in the blockmap file corresponds
to one block in the image file. Each entry counts how many
carved files contain this block. Requires more memory and
disk. **EXPERIMENTAL**
-n Don't add extensions to extracted files.
-o Set output directory for carved files.
-O Don't organize carved files by type. Default is to organize carved files
into subdirectories.
-p Perform image file preview; audit log indicates which files
would have been carved, but no files are actually carved.
-q Carve only when header is cluster-aligned.
-r Find only first of overlapping headers/footers [foremost 0.69 compat mode].
-s Skip n bytes in each disk image before carving.
-t Set directory for coverage blockmap. **EXPERIMENTAL**
-u Use carve coverage blockmap when carving. Carve only sections
of the image whose entries in the blockmap are 0. These areas
are treated as contiguous regions. **EXPERIMENTAL**
-V Print copyright information and exit.
-v Verbose mode.

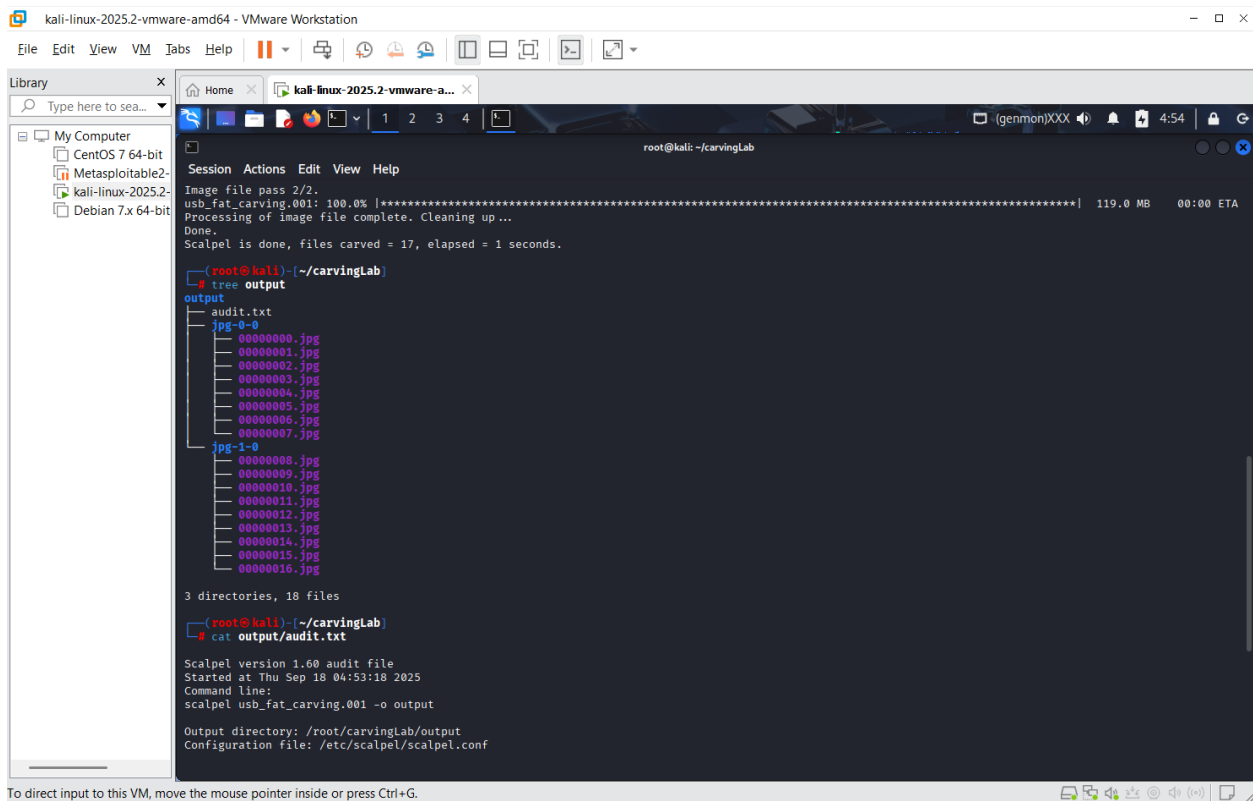
(root@kali) ~/carvingLab
# scalpel usb_fat_carving.001 -o output
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/root/carvingLab/usb_fat_carving.001"

Image file pass 1/2.
usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" -> 8 files
jpg with header "\xff\xd8\xff\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" -> 9 files
Carving files from image.
Image file pass 2/2.
usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 17, elapsed = 1 seconds.

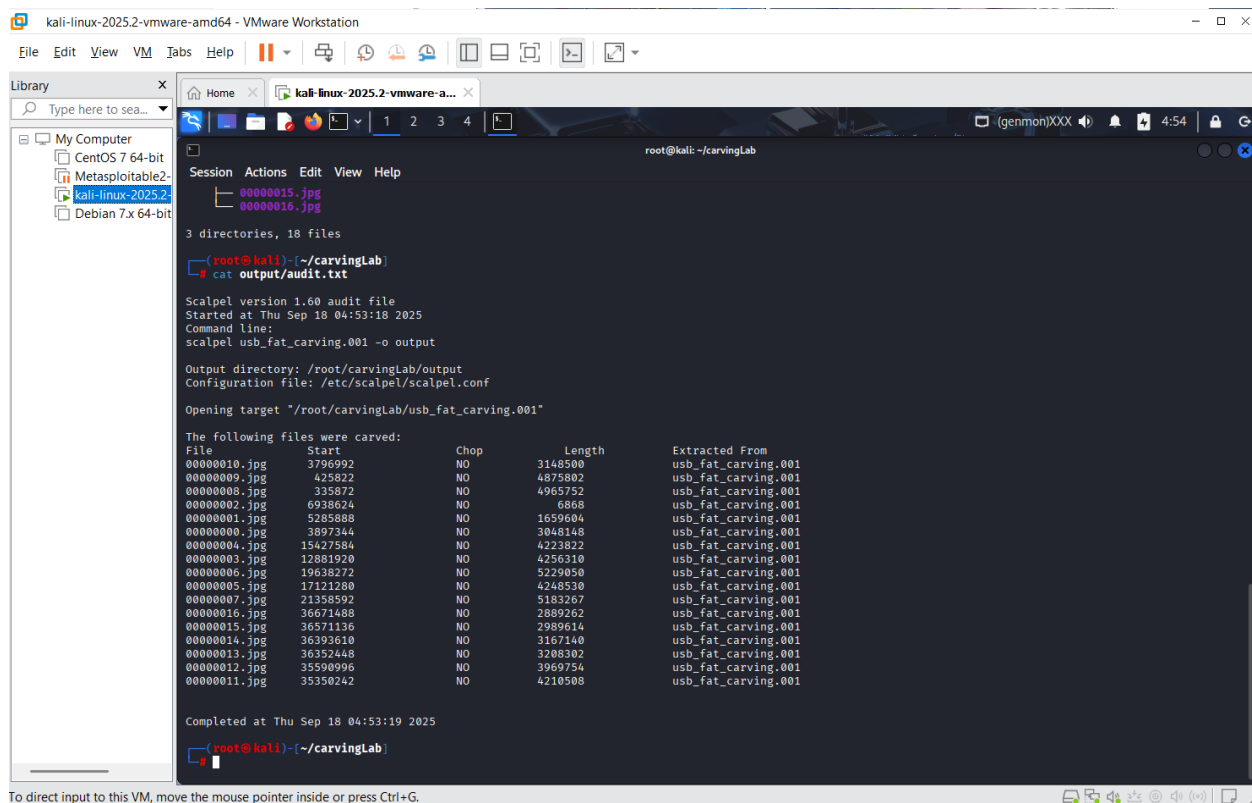
(root@kali) ~/carvingLab
```

ảnh trên: Carving the USB image



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

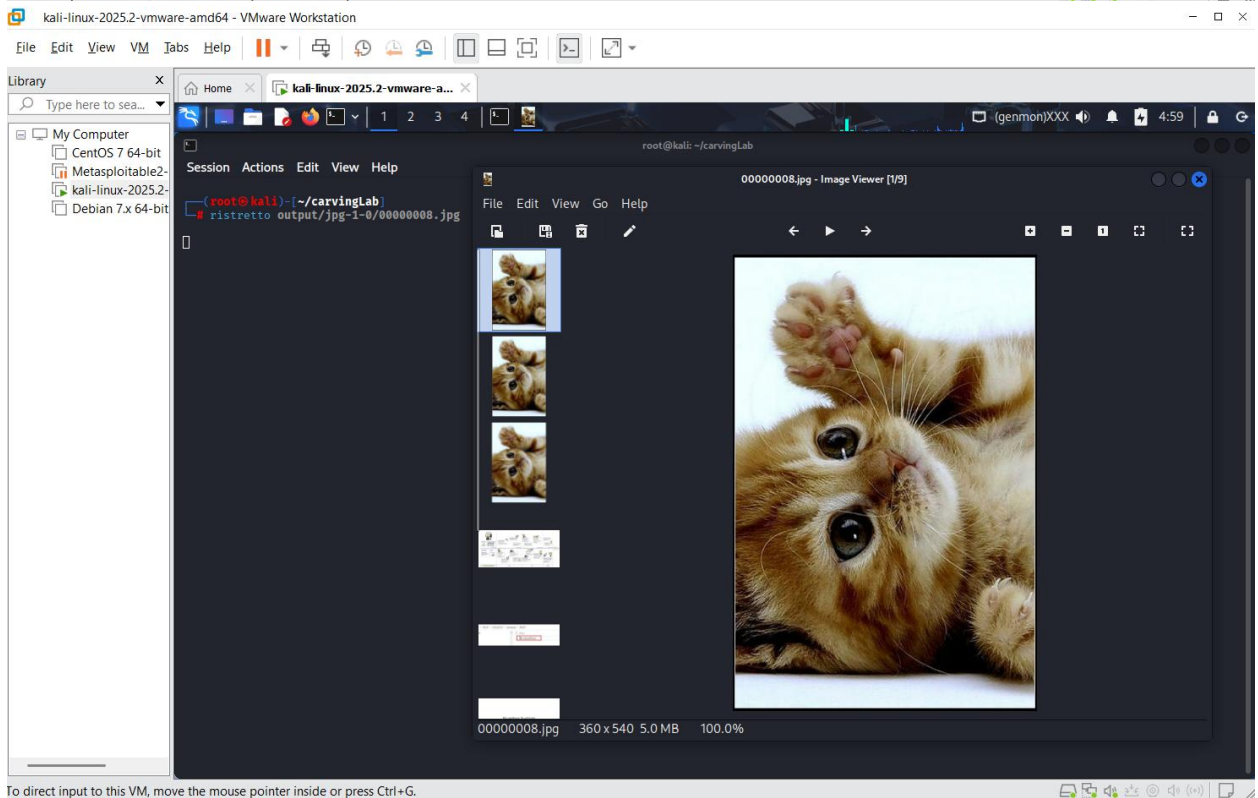
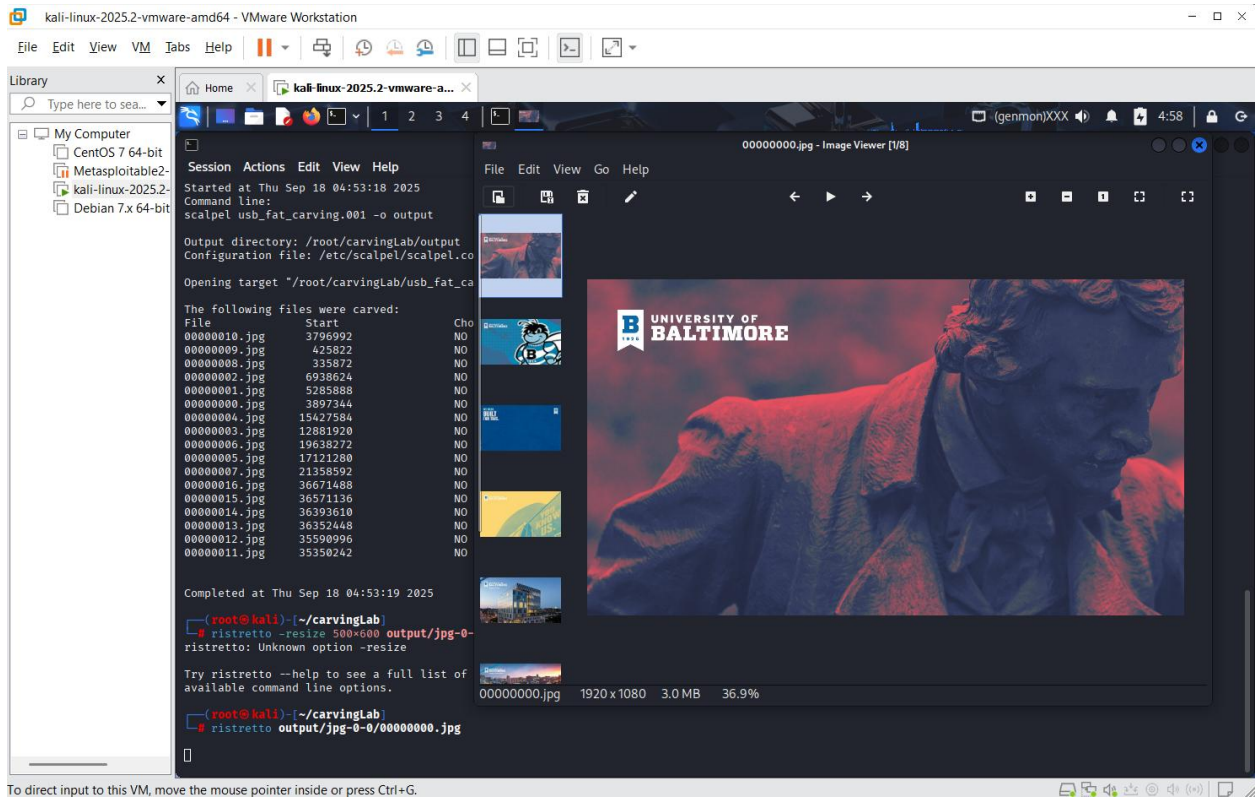
ảnh trên: hiển thị các tệp được carve



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

ảnh trên: Hiển thị nhật ký kiểm tra

Step 4



2 ảnh trên: 2 hình ảnh đã được khôi phục thành công